01 / START HERE

Build a bird habitat model

Which places offer suitable habitat for different birds? Use the supplied German data to build a model, then try it on anonymous locations in nearby Central Europe.

Saturday 24 October 2026, German time. Teams of 2-3 work from **11:00 to 17:30**. Presentations follow. The supplied starter runs on a CPU.

Your three tasks

- **Build a model for ten birds.** Give each requested bird-location pair a score: higher means more suitable according to your model.
- **Find and explain something useful.** Explore the bird data and show two or three clear findings with readable figures.
- **Optional extension:** test whether learning from other birds improves your model. It uses the same model and insight points, with no extra bonus or separate award.

How you earn points

Category	Points
Final model performance	Up to 50
Insights and visualizations	30
Presentation and questions	15
Reproducibility checklist	5

Match the shipped baseline: 15/50 model points. Improve its AP by 0.02, 0.05, 0.10 or 0.15 to earn **19.7, 26.7, 38.3 or 50**. AP measures how well reported presences rank above background. Higher is better.

Below the baseline, points rise from 0 at constant-score AP (1/6) to 15 at the reference; all model points stay between 0 and 50. The final comparison uses the same private split for your model and the fixed reference. The live AP board is provisional. See the Space Rules for the exact formula.

Competition hub and rules | Challenge data | Starter code

02 / YOUR INPUTS

Start with the small package

You can finish the modeling task with the 40 MB table package. Choose larger inputs only when they help your approach.

Download	Size	Use it for
Modeling tables	40 MB	Ready-to-use labels, environmental features, validation splits and a practice set. Start here.
Bird exploration	313 MB	About 2.49 million German observations across the available months, for your insights.
German image patches	20.22 GB	Two prepared spatial views around the modeling locations. Optional.
Whole German rasters	148.57 GB	The national AlphaEarth tiles. Optional.
Encrypted final inputs	4.30 MB tables; 3.10 GB with patches	Download before test unlock. Use the table or patch version that your model needs.

What is an AlphaEarth embedding?

Instead of an ordinary photograph, each 10 m ground cell has **64 numbers** that describe patterns learned by the AlphaEarth model. These numbers are inputs to your bird model. The supplied embeddings describe 2024.

The tables include the 10 m cell at each location, summaries of its surroundings within **100 m** and **1 km**, and data-coverage measures: 195 input columns in total. Those larger distances are circle radii, not coarser single pixels.

Patches keep the spatial arrangement: **128 x 128 cells at 10 m** cover 1.28 km across; **64 x 64 cells at 40 m** cover 2.56 km. A local pixel or patch can preserve a small city park that a broad neighborhood summary blends with nearby buildings.

> Use the starter loaders and **SPATIAL_INPUTS.md** for decoding, missing-data masks and the bird location inside each patch. These are 64-channel arrays, not RGB photos. Patches are north-up; whole tiles retain their source orientation.

The data collection contains 207 files, 172.24 GB total. Keep the folder structure, verify SHA256SUMS and use runtime-local storage. Code is separate. Retain the packaged licenses and attribution: record-specific CC0/CC BY for birds, CC BY 4.0 for AlphaEarth with Google/Google DeepMind credit, and BKG mask terms where used. Code licensing is separate.

03 / MODELING

Train and compare fairly

Everyone uses the same German validation set, so teams can compare their results on equal terms.

Target bird	German name
Eurasian Skylark	Feldlerche
Yellowhammer	Goldammer
Common Linnet	Bluthänfling
Red-backed Shrike	Neuntöter
Northern Lapwing	Kiebitz
Eurasian Nuthatch	Kleiber
Common Reed Warbler	Teichrohrsänger
Common Cuckoo	Kuckuck
Whinchat	Braunkehlchen
Lesser Spotted Woodpecker	Kleinspecht

Use the IDs and seasonal windows in species.csv. Eleven additional helper species have training labels ready for optional experiments.

What the labels mean

A positive label means the bird was reported. **A zero** is a sampled background example from the observation process; it does not prove the bird was absent. Unrecorded or unknown labels stay unknown. Your score is a relative ranking, not a verified probability of occupancy.

A reliable starting point

1. **Run the starter baseline.** Fit on the supplied training set and report AP on the common German development set - our validation set.
2. **Keep validation independent.** Fit every learned step, including scaling and helper models, only on allowed training rows. Use the supplied exclusions to keep nearby or related records out of training, across all birds.
3. **Compare one change at a time.** Five-fold spatial cross-validation is optional. After model choice, you may declare a final refit on training plus validation. Keep the earlier held-out result.

Reference: unchanged shipped starter, German training only. Validation and private AP differ. Fine pixels cannot fix uncertain coordinates. Results do not establish occupancy or causation.

04 / PREDICTIONS

Send your predictions

Your team runs the model and the Space checks and encrypts the result. A GitHub issue is the submission record that delivers your file to the evaluator.

1. **Register your team.** List 2-3 GitHub usernames and choose one captain to upload. After organizer approval, your short team ID (the team slug) appears on the Space.
2. **Try a practice delivery.** Select your team and Practice on the Space. Use the supplied 120-row practice CSV, encrypt it, download the JSON, attach it to the Practice issue form and create the issue.
3. **At 13:30, unlock the final inputs.** The Space provides the test key. Use the starter unlock tool on files already copied to your runtime. This key opens the input data.
4. **Predict all 35,510 rows.** That is ten birds at 3,551 locations. Export a CSV with exactly **row_id,score**. Match every row ID in the supplied sample once. Scores must be finite and between 0 and 1; the CSV limit is 5 MB. Row order does not matter.
5. **Upload your prediction.** On the Space, select Final-test prediction and check/encrypt your CSV. Download the **.encrypted.json**. Open the linked GitHub Prediction form, attach that JSON and wait for its file link. Then create the issue before **17:30**. Keep its issue number and wait for the evaluator's Accepted receipt. Downloading alone does not submit.
6. **Keep or change your final file.** Your **latest accepted prediction** is used by default, not your highest public score. Want an older accepted file? Open **Choose / confirm final entry**, enter its issue number and submit before 17:30. This form also records method/code and presentation references; it does not upload predictions again.

Per team: five accepted final predictions and ten accepted practice checks. Invalid files and identical retries do not consume an accepted attempt. Twenty invalid requests in a channel trigger its safety limit. Submission encryption is separate from the key that unlocks test inputs.

Example: #12 is accepted, then #16 is accepted. Do nothing and #16 is final. Choose #12 in the final-choice form and #12 is final instead. The latest valid choice wins. A rejected later upload does not replace your accepted file. Keep issues unchanged; corrections use a new issue. The GitHub issue creation time controls the deadline, even if scoring finishes later.

Final private scoring uses exactly one selected file per team. Full-precision AP and overall ties use its original receipt, then issue number. Overall-prize eligibility also requires supported visual insights, reproducibility references and a presentation.

05 / INSIGHTS AND PRESENTATION

Explain what you found

Choose a useful question and tell a short story supported by your analysis. Two or three clear figures are enough.

Good questions to explore

- Where and when are birds reported? Could uneven observation coverage explain the pattern? Compare feature scales or model errors with clear figures.
- For cross-species work, compare models with and without predicted helper signals on the same labels and validation. Observed helper labels are unavailable at final locations.

Prepare a five-minute presentation

Explain your question, data, model, common validation result and strongest findings. Finish with limitations. You have **five minutes plus two for questions**. A careful negative result is useful evidence.

Before 17:30, submit	Where and how
Your chosen prediction	Encrypted JSON in a GitHub Prediction issue. Use the final-choice form if you want an older accepted file.
Method and reproducible work	Complete the method report and share your code/checkpoint reference before 17:30; the final-choice form collects these links. Include features, seed, runtime, resource use, refit and the matching prediction hash.
Presentation	The captain uploads one PDF up to 10 MB through the presentation form, using the team slug, GitHub username and Google sign-in. A source presentation up to 100 MB is optional. Identify the response in your nomination.

Five evidence checks, one point each: code/configuration; reproduction commands and dependencies; the correct German validation result; features, preprocessing and helpers; accurate limitations. Complete the method report. A technical organizer checks the evidence.

Open the presentation upload form. Press Submit and keep the confirmation/time. Replace slides with a new complete response before 17:30; the latest valid one is used. Slides stay organizer-private.

Use supplied data and approved inputs. No extra bird observations, range maps, bird-distribution predictions or hidden labels. Extra environmental inputs need organizer approval and equal coverage for all teams. Do not refit preprocessing to anonymous tests.

06 / SATURDAY 24 OCTOBER

Plan your day

All times below are German local time: Europe/Berlin, CEST / MESZ (UTC+02:00).

Time	What happens
10:00	Registration and coffee. Code, guides and competition pages open. Check your setup.
10:15	Welcome and AI Nation introduction: Juliana Carneiro da Cunha Baumgartl, about 5 minutes.
10:25	Hula Earth and Ames explain the challenge and data, up to 10 minutes each.
10:45	Form teams, confirm your category and meet the mentors.
11:00	**Modeling and data analysis begin.**
13:00	Lunch in the room; no fixed work break.
13:30	**The test key appears on the Space.** Unlock your downloaded test files.
14:00	Mentors visit every team, including help with test inference and submissions.
16:30	Second mentor round. Focus on finishing predictions, figures and slides.
17:30	**All submissions close:** predictions, nomination, method/code reference and slides.
17:45-19:45	Presentations: 5 minutes speaking, 2 minutes questions.
19:45-20:00	Jury deliberation and final score checks.
20:00-20:15	Awards.
20:15-20:45	Dinner and networking.

Before 11:00: download/unpack files, read technical instructions, check accounts and test your environment with unrelated examples. Challenge-data exploration, features, modeling and visual analysis start at 11:00, including work done by AI agents.

Before 13:30: put the encrypted test files on your runtime disk. **At 16:30:** concentrate on finishing; technical fixes and submissions remain possible until 17:30. Keep a working baseline as a fallback and ask a mentor early if loading or delivery fails.

Final check: an accepted prediction, your final choice checked, the matching method/code reference and a submitted presentation PDF. The presentation period does not extend the submission deadline. Use the Space for links and event announcements.